

LOW ACRYLONITRILE NBR PATENT LANDSCAPE: EMULSION POLYMERIZATION, COMPOSITION CONTROL, AND WORK-UP TRENDS

1. ABSTRACT

This report synthesizes two primary patents relevant to nitrile butadiene rubber (NBR) manufacture, with emphasis on low-acrylonitrile and composition-controlled embodiments. The selected landscape spans one patent centered on controlled emulsion/block-copolymer technology that explicitly includes nitrile-rubber blocks containing acrylonitrile and butadiene, and one patent focused directly on nitrile rubbers produced by emulsion polymerization and subsequent coagulation/washing work-up. Across the set, the dominant synthesis mode is aqueous emulsion polymerization, often using surfactant systems, initiators such as potassium persulfate, and chain-transfer or molecular-weight-control agents such as tert-dodecyl mercaptan. The patents show two distinct but complementary trends: first, composition control during polymerization via staged or controlled monomer addition to tune acrylonitrile/butadiene incorporation; second, downstream isolation strategies that strongly affect storage stability and residual inorganic content, especially calcium and chlorine levels.

The NBR-focused patent discloses latices A, B, and C with acrylonitrile contents of 27, 12.5/6.5, and 44 parts by weight per 100 parts monomer mixture, respectively, and polymerization temperatures of 17, 18, and 20 °C. It further demonstrates that coagulation and washing conditions, including the use of calcium-containing water, influence storage stability and residual calcium/chlorine in the rubber. The block-copolymer patent is relevant because it explicitly defines a B block comprised of acrylonitrile/methacrylonitrile and isoprene or butadiene, and its examples show aqueous microemulsion polymerization with water, potassium persulfate, phosphate, and potassium hydroxide, followed by latex formation and stripping. Overall, the extracted evidence indicates that low-acrylonitrile NBR development is driven by emulsion-process control, monomer-feed staging, and careful post-polymerization work-up rather than by nonaqueous synthesis routes.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses controlled emulsion polymerization in an aqueous medium where an emulsifier is generated in situ from a latent surfactant and surfactant activator. The examples show microemulsion formation followed by rapid heating to produce stable latexes, and the claims extend the platform to block copolymers containing a nitrile-rubber B block made from acrylonitrile/methacrylonitrile and a conjugated diolefin such as butadiene or isoprene. - Latent surfactant: ~0.2 grams — Example 1 - Styrene: 5 milliliters — Example 1 - Surfactant activator in water: 5 milliliters — Example 1 - Styrene: 1,000 grams — Example 2 - Oleic acid: 60.0 grams — Example 2 - Dibenzyltrithiocarbonate: 7.2 grams — Example 2 - RO water: 4,000 grams — Example 2 - Potassium persulfate: 40.0 grams — Example 2 - Tripotassium phosphate: 40.0 grams — Example 2 - Potassium hydroxide: 16.4 grams — Example 2 - Reaction temperature: 65° C. — Example 2 - Conversion: complete conversion — Example 2 - Solids after stripping: 20.6% — Example 2 - Mn: 54,000 — Example 2 - PDI: 1.17 — Example 2 - Styrene: 1,000 grams — Example 3 - Hexadecyl half ester of maleic acid: 60.0 grams — Example 3 - Dibenzyltrithiocarbonate: 7.2 grams — Example 3 - RO water: 4,000 grams — Example 3 - Potassium persulfate: 40.0 grams — Example 3 - Tripotassium phosphate: 40.0 grams — Example 3 - Potassium hydroxide: 16.4 grams — Example 3 - Reaction temperature: 65° C. — Example 3 - Conversion: complete conversion — Example 3 - Solids after stripping: 23.7% — Example 3 - Mn: 48,000 — Example 3 - PDI: 1.17 — Example 3 - RO water: 200 grams — Example 4 - Seed PS latex: varying amounts — Example 4 - Isoprene: varying amounts — Example 4 - Butadiene: varying amounts — Example 4 - Butadiene/styrene mixture: 77/23 — Example 4 - A block: 5 weight percent to 30 weight percent of total block polymer — Claim 5 - B block: 70 weight percent to 95 weight percent of total block polymer — Claim 5 - A block: 12 weight percent to 25 weight percent of total block polymer — Claim 6 - B block: 75 weight percent to 88 weight percent of total block polymer — Claim 6 - Styrene: 1,000 grams — Example 6 - Oleic acid: 60.0 grams — Example 6 - PBN: 10.0 grams — Example 6 - RO water: 4,000 grams — Example 6 - Potassium persulfate: 40.0 grams — Example 6 - Tripotassium phosphate: 40.0 grams — Example 6 - Potassium hydroxide: 16.4 grams — Example 6 - Reaction temperature: 75° C. — Example 6 - Solids after stripping: 22.7% — Example 6 - Mn: 25,000 — Example 6 - PDI: 1.80 — Example 6 - RO water: 241 grams — Example 5 - RO water: 197.1 grams — Example 5 - RO water: 84.7 grams — Example 5 - RO water: 160.5 grams — Example 5 - RO water: 200 grams — Example 8 - Distilled water: 200 grams — Example 8 - Potassium persulfate: 2.06 grams — Example 8 - Tripotassium phosphate: 2.13 grams — Example 8 -

Potassium hydroxide: 0.9 grams — Example 8 - Reaction temperature: 65° C. — Example 8 - Mn: 35,000 — Example 8 - Mw: 61,000 — Example 8 - PDI: 1.74 — Example 8 - Styrene: 200 grams — Example 10 - Oleic acid: 12.0 grams — Example 10 - PBN: 2.0 grams — Example 10 - Distilled water: 500 grams — Example 10 - Potassium persulfate: 8.0 grams — Example 10 - Tripotassium phosphate: 8.0 grams — Example 10 - Potassium hydroxide: 3.6 grams — Example 10 - Reaction temperature: 75° C. — Example 10 - Mn: 39,000 — Example 10 - Mw: 48,000 — Example 10 - PDI: 1.22 — Example 10 - PIN: equimolar amount — Example 11 - Completion time: about 7 hours — Example 11 - Mn: 41,000 — Example 11 - Mw: 70,000 — Example 11 - PDI: 1.70 — Example 11 - Latex from Example 10: 100 grams of about 22% solids — Example 12 - Potassium persulfate: 0.5 grams — Example 12 - Distilled water: 100 milliliters — Example 12 - n-Butyl acrylate: 22.0 grams — Example 12 - Reaction temperature: about 65° C. — Example 12 - Weak exotherm: about 70° C. — Example 12 - Completion time: about 2 hours — Example 12 - Mn: 45,000 — Example 12 - Mw: 77,000 — Example 12 - PDI: 1.70 — Example 12 - Oleic acid: 3.0 grams — Example 13 - Benzaldehyde: 0.31 grams — Example 13 - 15% aqueous solution of N-isopropylhydroxylamine: 1.4 grams — Example 13 - Stirring time: 1 hour — Example 13 - Distilled water: 200 grams — Example 13 - Potassium persulfate: 2.06 grams — Example 13 - Tripotassium phosphate: 2.13 grams — Example 13 - Potassium hydroxide: 0.9 grams — Example 13 - Reaction temperature: 75° C. — Example 13 - Completion time: about 4 hours — Example 13 - Mn: 31,000 — Example 13 - Mw: 55,000 — Example 13 - PDI: 1.76 — Example 13 - Styrene: 50.0 grams — Example 14 - Dodecylbenzenesulfonic acid: 3.0 grams — Example 14 - Distilled water: 1 milliliter — Example 14 - Sodium nitrite: 0.171 grams — Example 14 - Distilled water: 200 grams — Example 14 - Potassium persulfate: 2.06 grams — Example 14 - Tripotassium phosphate: 2.13 grams — Example 14 - Potassium hydroxide: 0.74 grams — Example 14 - Reaction temperature: 75° C. — Example 14 - Completion time: about 3 hours — Example 14 - Mn: 97,000 — Example 14 - Mw: 147,000 — Example 14 - PDI: 1.51 — Example 14 - Styrene: 50 grams — Example 15 - Oleic acid: 3.0 grams — Example 15 - DPE: 0.448 grams — Example 15 - Distilled water: 200 grams — Example 15 - Potassium persulfate: 2.06 grams — Example 15 - Tripotassium phosphate: 2.13 grams — Example 15 - Potassium hydroxide: 0.92 grams — Example 15 - Reaction temperature: 75° C. — Example 15 - Completion time: about 2 hours — Example 15 - Mn: 51,000 — Example 15 - Mw: 92,000 — Example 15 - PDI: 1.80 — Example 15 - Styrene: 2,240 grams — Example 16 - Oleic acid: 134.4 grams — Example 16 - Dibenzyltrithiocarbonate: 16.0 grams — Example 16 - RO water: 3,808 grams — Example 16 - Potassium persulfate: 13.44 grams — Example 16 - Tripotassium phosphate: 89.6 grams — Example 16 - Potassium hydroxide: 36.96 grams — Example 16 - Reaction temperature: 65° C. — Example 16 - Solids: 38.9% — Example 16 - Completion time: less than 3 hours — Example 16 - Mn: 45,500 — Example 16 - PDI: 1.11 — Example 16 - Sodium nitrite: 0.69 grams — Example 17 - Oleic acid: 3.0 grams — Example 17 - Styrene: 50 grams — Example 17 - Concentrated acetic acid: 1.0 grams — Example 17 - Water: 2 grams — Example 17 - Distilled water: 160 grams — Example 17 - Potassium persulfate: 2.0 grams — Example 17 - Tripotassium phosphate: 2.0 grams — Example 17 - Potassium hydroxide: 2.35 grams — Example 17 - Reaction temperature: 75° C. — Example 17 - Conversion: 26.1% solids — Example 17 - Completion time: about 2.5 hours — Example 17 - Mn: 96,300 — Example 17 - Mw: 135,000 — Example 17 - PDI: 1.4 — Example 17 - Styrene: 50 grams — Example 18 - Oleic acid: 3.0 grams — Example 18 - 1-chloro-1,1-diphenylethane: approximately 0.01 moles — Example 18 - Concentrated HCl: 10

milliliters — Example 18 - Toluene: 5 milliliters — Example 18 - Distilled water: 200 grams — Example 18 - Potassium persulfate: 2.0 grams — Example 18 - Tripotassium phosphate: 2.0 grams — Example 18 - Potassium hydroxide: 1.55 grams — Example 18 - Reaction temperature: 75° C. — Example 18 - Completion time: about 4 hours — Example 18 - Solids: 22.7% — Example 18 - Mn: 40,000 — Example 18 - PDI: 1.37 — Example 18 - Styrene: 1,300 grams — Example 19 - Oleic acid: 78.0 grams — Example 19 - Crude 1-chloro-1,1-diphenylethane: 39.0 grams — Example 19 - Deionized reverse osmosis water: 2,470 grams — Example 19 - Potassium persulfate: 50.7 grams — Example 19 - Tripotassium phosphate: 50.7 grams — Example 19 - Potassium hydroxide: 29.5 grams — Example 19 - Reaction temperature: 75° C. — Example 19 - Solids: 38.6% — Example 19 - Completion time: about 5.5 hours — Example 19 - Mn: 32,000 — Example 19 - PDI: 1.85 — Example 19 - 1-hexene: 25 grams — Example 20 - n-butyl acrylate: 25 grams — Example 20 - Dipyrityl: 0.38 grams — Example 20 - Oleic acid: 4.0 grams — Example 20 - Ethyl 2-bromoisobutyrate: 0.47 grams — Example 20 - KOH: 1.7 grams — Example 20 - Distilled water: 139 milliliters — Example 20 - Copper sulfate pentahydrate solution: 60.9 grams of 1% aqueous solution — Example 20 - Hydrazine hydrate: 2 drops — Example 20 - Reaction temperature: about 67° C. — Example 20 - Reaction time: 1.5 hours — Example 20 - Conversion: 10.9% solids — Example 20 - ¹³C-NMR incorporation: 5.5 mole % 1-hexene — Example 20 - Medium & Water Role: Aqueous emulsion/microemulsion polymerization is the core reaction medium, with water explicitly used in the polymerization feed and latex formation; later water is also used in seed-latex operations and stripping/workup. The evidence does not show a nonaqueous NBR synthesis route. - acrylonitrile content: Not disclosed in extracted evidence

Medium & Water Role - Aqueous emulsion/microemulsion polymerization is the core reaction medium, with water explicitly used in the polymerization feed and latex formation; later water is also used in seed-latex operations and stripping/workup. The evidence does not show a nonaqueous NBR synthesis route. - Core reaction medium: Aqueous emulsion / microemulsion polymerization medium

acrylonitrile content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Demonstrated in-situ emulsification using latent surfactant/surfactant activator pairs in styrene/water. - Example 2: Showed RAFT-controlled emulsion polymerization in water to give polystyrene latex with low PDI. - Example 3: Showed in-situ generated polymerizable surfactant enabling controlled emulsion polymerization and stable latex formation. - Example 4: Demonstrated seed-latex emulsion triblock copolymer formation using isoprene, butadiene, or butadiene/styrene mixtures. - Example 5: Demonstrated preparation of S-NBR-S triblock copolymer latexes from PS seed latexes.

Technical Relevance This patent is relevant because it explicitly claims block polymers containing a nitrile-rubber B block made from acrylonitrile/methacrylonitrile and a conjugated diolefin, which overlaps with NBR chemistry even though the examples are largely styrene-based. It also provides an aqueous controlled-emulsion platform and seed-latex methodology that are directly applicable to composition-controlled nitrile-rubber synthesis.

Patent 2

- Patent Number: US7923518B2
- Patent Title: Nitrile rubbers
- Assignee: Lanxess Deutschland Gmbh
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion polymerization of nitrile rubbers in aqueous latex form using butadiene and acrylonitrile, with controlled addition of monomer and chain-transfer agent. The latices are then coagulated and washed under varied conditions to produce solid nitrile rubber with improved storage stability and controlled residual calcium/chlorine content. - Butadiene: 73 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Acrylonitrile: 27 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Total amount of water: 190 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Erkantol BXG: 3.69 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Baykanol PQ: 1.10 parts by weight per 100 parts monomer mixture — Table 1, Latex A - K salt of coconut fatty acid: 0.73 parts by weight per 100 parts monomer mixture — Table 1, Latex A - KOH: 0.05 parts by weight per 100 parts monomer mixture — Table 1, Latex A - t-DDM: 0.24/0.24 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Potassium peroxodisulphate: 0.39/0.19 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Tris(α -hydroxyethyl)amine: 0.57 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Na dithionite: 1.20 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Potassium hydroxide: 1.28 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Vulkanox KB: 1.25 parts by weight per 100 parts monomer mixture — Table 1, Latex A - Polymerization temperature: 17 °C — Table 1, Latex A - Polymerization conversion: 75 % — Table 1, Latex A - Polymerization time: 11 h — Table 1, Latex A - Butadiene: 81 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Acrylonitrile: 12.5/6.5 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Total amount of water: 180 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Erkantol BXG: 3.69 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Baykanol PQ: 1.10 parts by weight per 100 parts monomer mixture — Table 1, Latex B - K salt of coconut fatty acid: 0.73 parts by weight per 100 parts monomer mixture — Table 1, Latex B - KOH: 0.05 parts by weight per 100 parts monomer mixture — Table 1, Latex B - t-DDM: 0.27/0.15/0.06 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Potassium peroxodisulphate: 0.45/0.20 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Tris(α -hydroxyethyl)amine: 0.61 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Na dithionite: 1.20 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Potassium hydroxide: 1.28 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Vulkanox KB: 1.25 parts by weight per 100 parts monomer mixture — Table 1, Latex B - Polymerization temperature: 18 °C — Table 1, Latex B - Polymerization conversion: 75 % — Table 1, Latex B - Polymerization time: 15 h — Table 1, Latex B - Butadiene: 56 parts by weight per 100 parts monomer mixture — Table 1, Latex C - Acrylonitrile: 44 parts by

weight per 100 parts monomer mixture — Table 1, Latex C - Total amount of water: 170 parts by weight per 100 parts monomer mixture — Table 1, Latex C - Erkantol BXG: 3.69 parts by weight per 100 parts monomer mixture — Table 1, Latex C - Baykanol PQ: 1.10 parts by weight per 100 parts monomer mixture — Table 1, Latex C - K salt of coconut fatty acid: 0.73 parts by weight per 100 parts monomer mixture — Table 1, Latex C - KOH: 0.05 parts by weight per 100 parts monomer mixture — Table 1, Latex C - t-DDM: 0.5/0.2 parts by weight per 100 parts monomer mixture — Table 1, Latex C - Potassium peroxodisulphate: 0.27 parts by weight per 100 parts monomer mixture — Table 1, Latex C - Tris(α -hydroxyethyl)amine: 0.15 parts by weight per 100 parts monomer mixture — Table 1, Latex C - Na dithionite: 1.20 parts by weight per 100 parts monomer mixture — Table 1, Latex C - Potassium hydroxide: 1.28 parts by weight per 100 parts monomer mixture — Table 1, Latex C - Vulkanox KB: 1.25 parts by weight per 100 parts monomer mixture — Table 1, Latex C - Polymerization temperature: 20 °C — Table 1, Latex C - Polymerization conversion: 74.5 % — Table 1, Latex C - Polymerization time: 7.5 h — Table 1, Latex C - Monomer mixture: 350 kg — batchwise autoclave process - Total amount of water: 700 kg — batchwise autoclave process - Initial water charge: 650 kg — batchwise autoclave process - Acrylonitrile: 12.5 parts by weight initially, 6.5 parts by weight at 35% conversion — Latex B description - t-DDM: 0.27 part by weight initially, 0.15 part by weight at 20% conversion, 0.06 part by weight at 40% conversion — Latex B description - t-DDM: 0.5 part by weight initially, 0.2 part by weight at 15% conversion — Latex C description - Coagulation/washing temperature: 20 °C — Examples 1-10 - Washing time: 3.8 h — Example 1 - Washing time: 3.5 h — Example 2 - Washing time: 3.5 h — Example 3 - Washing time: 20 h — Example 11 - Washing time: 7.75 h — Example 11 - Washing time: 5 h — Example 13 - Washing time: 5 h — Example 14 - Washing time: 5 h — Example 15 - Washing time: 5 h — Example 17 - Washing time: 2.5 h — Example 19 - Washing time: 2.5 h — Example 20 - Storage stability: not more than 3 Mooney units — Claim 1 - Calcium content: 150 ppm to 1290 ppm — Claim 1 - Chlorine content: at least 40 ppm — Claim 1 - Calcium content: 200 ppm to 1290 ppm — Claim 2 - Calcium content: 400 ppm to 1290 ppm — Claim 3 - Calcium content: 500 ppm to 1290 ppm — Claim 4 - Calcium content: 600 ppm to 1290 ppm — Claim 5 - Calcium content: 800 ppm to 1290 ppm — Claim 6 - Acrylonitrile content: 28.9% by weight — Examples 1, 11, 21 - Acrylonitrile content: 18.8% by weight — Example 32 - Acrylonitrile content: 38.6% by weight — Example 36 - Calcium content: 1235 to 1290 ppm — Examples 1-10 discussion - Calcium content: 490 to 890 ppm — Examples 11-20 discussion - Medium & Water Role: The core synthesis is aqueous emulsion latex polymerization of butadiene/acrylonitrile, and water is also used extensively in coagulation, crumb washing, and steam distillation/workup. The evidence clearly distinguishes the polymerization medium from downstream water-based isolation steps. - acrylonitrile content: 12.5/6.5 parts by weight per 100 parts by weight of the monomer mixture — Table 1, Latex B

Medium & Water Role - The core synthesis is aqueous emulsion latex polymerization of butadiene/acrylonitrile, and water is also used extensively in coagulation, crumb washing, and steam distillation/workup. The evidence clearly distinguishes the polymerization medium from downstream water-based isolation steps. - Core reaction medium: Aqueous emulsion latex polymerization

acrylonitrile content - 12.5/6.5 parts by weight per 100 parts by weight of the monomer mixture
— Table 1, Latex B

Relevant Experimental Evidence - Example series 1-5: Showed that coagulation with calcium chloride and washing conditions strongly affect storage stability and residual calcium/chlorine in NBR from latex A. - Example series 2: Showed that using undeionized, calcium-containing water in crumb washing gives storage-stable nitrile rubbers, unlike deionized water. - Example series 3: Showed that varying coagulation and washing conditions can produce storage-stable nitrile rubbers. - Latex A/B/C preparation: Demonstrated emulsion polymerization recipes with staged monomer and t-DDM addition at 17-20 °C. - Examples 32 and 36: Extended the storage-stability/coagulation findings to NBRs with 18.8 wt% and 38.6 wt% acrylonitrile.

Technical Relevance This patent is directly relevant because it centers on nitrile rubber manufacture from butadiene and acrylonitrile in emulsion, including a low-acrylonitrile latex B formulation with 12.5/6.5 parts acrylonitrile and staged feed addition. It also provides practical work-up guidance showing how coagulation and washing conditions influence storage stability and residual calcium/chlorine, which are important downstream quality attributes for low-ACN NBR.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	acrylonitrile content	Key Findings
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US7592409B2	The Goodyear Tire & Rubber Company	2009	Aqueous emulsion/microemulsion polymerization is the core reaction medium, with water explicitly used in the polymerization feed and latex formation; later water is also used in seed-latex operations and stripping/workup. The evidence does not show a nonaqueous NBR synthesis route.	Not disclosed in extracted evidence	Example 1: Demonstrated in-situ emulsification using latent surfactant/surfactant activator pairs in styrene/water.
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US7923518B2	Lanxess Deutschland GmbH	2011	The core synthesis is aqueous emulsion latex polymerization of butadiene/acrylonitrile, and water is also used extensively in coagulation, crumb washing, and steam distillation/workup. The evidence clearly distinguishes the polymerization medium from downstream water-based isolation steps.	12.5/6.5 parts by weight per 100 parts by weight of the monomer mixture — Table 1, Latex B	Example series 1-5: Showed that coagulation with calcium chloride and washing conditions strongly affect storage stability
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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7592409B2 is a general controlled emulsion/block-copolymer platform that only indirectly touches nitrile-rubber chemistry through a claimed B block containing acrylonitrile/methacrylonitrile and butadiene/isoprene; it does not disclose an NBR composition value for the target material.
- US7923518B2 is the only patent in the set that directly discloses NBR recipes with explicit acrylonitrile loadings, including a low-ACN latex B formulation using 12.5/6.5 parts acrylonitrile and staged feed addition.

- Both patents use aqueous emulsion/microemulsion media, but US7592409B2 emphasizes in-situ surfactant generation and controlled radical/block architecture, whereas US7923518B2 emphasizes conventional NBR emulsion polymerization plus coagulation/washing work-up.
- US7923518B2 discloses much more process detail for NBR manufacture: monomer ratios, emulsifier package, chain-transfer agent t-DDM, initiator system, temperatures of 17-20 °C, and conversion levels around 74.5-75%.
- Only US7923518B2 provides downstream isolation data tied to product quality, including calcium content, chlorine content, washing temperature at 20 °C, and storage stability not more than 3 Mooney units.

4. CONCLUSION

Across the selected primary patents, the strongest direct evidence for low-acrylonitrile NBR synthesis comes from US7923518B2, which discloses butadiene/acrylonitrile emulsion recipes including a low-ACN staged-feed latex B formulation and links composition to storage-stable rubber after coagulation and washing. US7592409B2 is relevant mainly as an enabling aqueous controlled-emulsion/block-copolymer platform that includes nitrile-rubber block chemistry, but it does not directly disclose a low-ACN NBR target composition. Overall, the extracted record shows that low-acrylonitrile NBR development is driven by emulsion polymerization recipe design, staged monomer addition, and careful post-polymerization work-up.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | The Goodyear Tire & Rubber Company | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US7923518B2 | Nitrile rubbers | Lanxess Deutschland GmbH | US | 2011 | <https://patents.google.com/patent/US7923518B2/en>